

Confidence-Aware, Explainable Multi-Species Bird Presence Detection Using Bioacoustic Signals

Technical Report: Algorithm & Methodology

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Date	February 17, 2026
Version	2.0
Model	EfficientNet-B0
Species Classified	54
Best Accuracy	71.32% (Validation)

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1. Executive Summary

This report presents a comprehensive technical analysis of a deep learning system designed for automatic bird species identification from audio recordings. The system employs a Convolutional Neural Network (CNN) based on the EfficientNet-B0 architecture, utilizing transfer learning from ImageNet and fine-tuned on mel-spectrogram representations of bird vocalizations.

Key Technical Components:

- **Algorithm:** Deep Convolutional Neural Network (CNN)
- **Architecture:** EfficientNet-B0 with custom classification head
- **Feature Representation:** Mel-spectrograms (128 mel bands × 216 time frames)
- **Optimization:** AdamW optimizer with cosine annealing learning rate schedule
- **Loss Function:** Cross-Entropy with label smoothing (and optional Focal Loss)
- **Augmentation:** SpecAugment + Mixup for improved generalization

Performance Results:

Metric	Value	Description
Test Accuracy	67.42%	Correct predictions on held-out test set
Top-5 Accuracy	82.57%	Correct species in top 5 predictions
Macro F1	58.39%	Average F1 across all 54 species
Weighted F1	66.42%	Sample-weighted average F1
Parameters	4.35M	Total trainable parameters
Training Time	~30 min	With pre-computed spectrograms

Table 1: Key Performance Metrics

2. Problem Statement

Objective: Develop an automated system capable of identifying bird species from environmental audio recordings with high accuracy and reliability.

Formal Definition:

Given an audio recording x of duration T seconds, the task is to predict the bird species y from a set of $K = 54$ possible species. This is formulated as a multi-class classification problem:

$$\hat{y} = \underset{k \in \{1, \dots, K\}}{\operatorname{argmax}} P(y=k | x; \theta)$$

where θ represents the learned model parameters and $P(y=k|x;\theta)$ is the predicted probability for species k .

2.1 Challenges

- **Acoustic Variability:** Bird calls vary by individual, age, season, and location
- **Environmental Noise:** Recordings contain wind, rain, insects, and other birds
- **Class Imbalance:** Some species have 1400+ samples, others have fewer than 50
- **Overlapping Calls:** Multiple species may vocalize simultaneously
- **Similar Vocalizations:** Some species produce acoustically similar calls

3. Algorithm Overview

The bird species classification system employs a multi-stage pipeline combining signal processing, deep learning, and transfer learning techniques. The core algorithm is a **Convolutional Neural Network (CNN)** that processes visual representations of audio.

3.1 Pipeline Architecture

Stage	Component	Algorithm/Technique	Output
1	Audio Loading	Digital Signal Processing	Waveform (22050 Hz)
2	Segmentation	Sliding Window	5-second chunks
3	Feature Extraction	STFT + Mel Filter Bank	Mel-spectrogram
4	Normalization	Log Compression + MinMax	Normalized image
5	CNN Backbone	EfficientNet-B0	1280-dim features
6	Classification	Dense + Softmax	54 probabilities

Table 2: Processing Pipeline Stages

3.2 Algorithm Classification

- **Supervised Learning:** Model learns from labeled examples (audio → species)
- **Deep Learning:** Neural networks with multiple hidden layers
- **Convolutional Neural Networks:** Exploits spatial patterns in spectrograms
- **Transfer Learning:** Leverages pre-trained ImageNet weights

4. Feature Extraction: Audio to Spectrogram

The first critical step is converting raw audio waveforms into visual representations that CNNs can process effectively. This is achieved through mel-spectrogram computation.

4.1 Short-Time Fourier Transform (STFT)

The STFT decomposes an audio signal into its frequency components over time. For a discrete signal $x[n]$, the STFT computes the Fourier transform over short overlapping windows:

$$X[m, k] = \sum x[n + mH] \cdot w[n] \cdot e^{-j2\pi kn/N}$$

Where: m = frame index, k = frequency bin, N = FFT size (2048), H = hop length (512), $w[n]$ = Hann window function.

4.2 Mel Filter Bank

The mel scale approximates human auditory perception. The conversion from frequency f (Hz) to mel scale m is:

$$m = 2595 \cdot \log_{10}(1 + f/700)$$

4.3 Feature Extraction Parameters

Parameter	Value	Purpose
Sample Rate	22,050 Hz	Standard audio rate
FFT Size	2048	Frequency resolution (~10.7 Hz/bin)
Hop Length	512	Time resolution (~23 ms/frame)
Mel Bands	128	Frequency bins in mel scale
Freq Min	150 Hz	Lower bound
Freq Max	15,000 Hz	Upper bound
Duration	5 seconds	Input segment length
Output Shape	(128, 216)	Mel bands × time frames

Table 3: Mel-Spectrogram Parameters

5. Model Architecture: EfficientNet-B0

EfficientNet-B0 is a convolutional neural network architecture developed by Tan and Le (2019) that achieves state-of-the-art accuracy while being significantly more parameter-efficient than previous architectures like ResNet or VGG.

5.1 Key Innovations

- **Compound Scaling:** Uniformly scales depth, width, and resolution
- **Mobile Inverted Bottleneck (MBConv):** Efficient building block
- **Squeeze-and-Excitation (SE):** Channel-wise attention mechanism
- **Swish Activation:** $f(x) = x \cdot \sigma(x)$, smoother than ReLU

5.2 Network Architecture

Stage	Operator	Resolution	Channels	Layers
1	Conv3×3	112×108	32	1
2	MBConv1, k3×3	112×108	16	1
3	MBConv6, k3×3	56×54	24	2
4	MBConv6, k5×5	28×27	40	2
5	MBConv6, k3×3	14×14	80	3
6	MBConv6, k5×5	14×14	112	3
7	MBConv6, k5×5	7×7	192	4
8	MBConv6, k3×3	7×7	320	1
9	Conv1×1 & Pool	1×1	1280	1
10	Dense + Softmax	-	54	1

Table 4: EfficientNet-B0 Architecture

5.3 Mobile Inverted Bottleneck (MBConv)

The MBConv block is the fundamental building unit consisting of:

1. **Expansion:** 1×1 convolution to increase channels by factor of 6
2. **Depthwise Conv:** Spatial filtering per channel (k×k kernel)
3. **Squeeze-Excitation:** Channel attention via global pooling + FC layers
4. **Projection:** 1×1 convolution to reduce channels
5. **Skip Connection:** Residual connection when dimensions match

5.4 Custom Classification Head

Layer	Input	Output	Activation	Regularization
Global Avg Pool	(1280,H,W)	1280	-	-
Dropout	1280	1280	-	p=0.5
Dense	1280	256	ReLU	-
BatchNorm	256	256	-	Normalization
Dropout	256	256	-	p=0.3
Dense	256	54	Softmax	-

Table 5: Custom Classification Head

6. Transfer Learning Approach

Transfer learning enables leveraging knowledge from a source domain (ImageNet image classification) to improve performance on our target domain (bird audio classification).

6.1 Why Transfer Learning Works

Despite the domain difference between natural images and audio spectrograms, transfer learning is effective because:

- 1. Low-level Feature Sharing:** Early CNN layers learn universal features (edges, textures, gradients) that transfer well across domains.
- 2. Hierarchical Representations:** The network learns increasingly abstract features useful for both image and spectrogram analysis.
- 3. Better Initialization:** Pre-trained weights provide a superior starting point compared to random initialization, enabling faster convergence.

6.2 Fine-tuning Strategy

Approach	Layers Trained	When to Use	Our Choice
Feature Extraction	Only classifier	Very limited data	✗
Partial Fine-tuning	Top layers only	Limited data	✗
Full Fine-tuning	All layers	Sufficient data	✓

Table 6: Transfer Learning Strategies

7. Optimization Algorithm: AdamW

AdamW (Adam with Decoupled Weight Decay) combines momentum-based optimization with adaptive learning rates and proper L2 regularization.

7.1 Algorithm Steps

For each parameter θ at iteration t :

1. Compute gradient: $g_t = \nabla L(\theta_{t-1})$
2. Update first moment: $m_t = \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$
3. Update second moment: $v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$
4. Bias correction: $\hat{m}_t = m_t / (1 - \beta_1^t)$, $\hat{v}_t = v_t / (1 - \beta_2^t)$
5. Update: $\theta_t = \theta_{t-1} - \alpha \cdot (\hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon) + \lambda \cdot \theta_{t-1})$

7.2 Hyperparameters

Parameter	Symbol	Value	Purpose
Learning Rate	α	0.0003	Step size
Beta 1	β_1	0.9	First moment decay
Beta 2	β_2	0.999	Second moment decay
Epsilon	ϵ	1e-8	Numerical stability
Weight Decay	λ	0.05	L2 regularization

Table 7: AdamW Hyperparameters

7.3 Learning Rate Schedule

We use **Cosine Annealing with Warm Restarts**, which periodically resets the learning rate to escape local minima. The schedule follows:

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min})(1 + \cos(T_{cur}/T_i \cdot \pi))$$

8. Loss Functions

8.1 Cross-Entropy Loss

The standard cross-entropy loss for multi-class classification:

$$L_{CE} = -\sum y_k \cdot \log(p_k)$$

8.2 Label Smoothing

Label smoothing prevents overconfident predictions by softening targets:

$$y_k^{smooth} = (1 - \epsilon) \cdot y_k + \epsilon / K$$

With $\epsilon = 0.1$ and $K = 54$ classes:

- Original: [0, 0, 1, 0, ..., 0] (one-hot)
- Smoothed: [0.0019, 0.0019, 0.9019, 0.0019, ...]

8.3 Focal Loss (For Class Imbalance)

Focal Loss down-weights easy examples, focusing on hard cases:

$$L_{FL} = -\alpha_t (1 - p_t)^\gamma \cdot \log(p_t)$$

Where $\gamma = 2.0$ (focusing parameter). When $\gamma = 0$, focal loss equals cross-entropy.

Loss Function	Handles Imbalance	Prevents Overconfidence
Cross-Entropy	✗	✗
CE + Label Smoothing	✗	✓
Focal Loss	✓	✗
Focal + Smoothing	✓	✓

Table 8: Loss Function Comparison

9. Data Augmentation Techniques

Data augmentation increases training diversity, improving generalization.

9.1 SpecAugment

SpecAugment (Park et al., 2019) applies masking directly to spectrograms:

Frequency Masking: Mask F consecutive mel bands

Time Masking: Mask T consecutive time frames

Parameter	Value	Description
Freq Mask (F)	20 bands	Max frequency bands to mask
Time Mask (T)	30 frames	Max time frames to mask
Num Masks	2 each	Masks per sample
Probability	0.5	Application probability

Table 9: SpecAugment Parameters

9.2 Mixup

Mixup (Zhang et al., 2018) creates virtual examples by interpolating pairs:

$$\begin{aligned}x_{\lambda} &= \lambda x_i + (1-\lambda)x_j \\ y_{\lambda} &= \lambda y_i + (1-\lambda)y_j\end{aligned}$$

Where $\lambda \sim \text{Beta}(0.4, 0.4)$. This encourages linear behavior between examples.

9.3 Audio Augmentations

Augmentation	Range	Effect
Time Shift	±15%	Circular shift in time
Pitch Shift	±2 semitones	Change pitch
Volume	0.7x - 1.3x	Simulate distance
Noise	SNR 15-30 dB	Add Gaussian noise

Table 10: Audio Augmentations

10. Training Methodology

10.1 Dataset

Metric	Value
Total Recordings	1,926
Total Chunks	34,811 (5-second segments)
Species	54 North American birds
Train Set	23,630 chunks (70%)
Validation Set	4,718 chunks (15%)
Test Set	5,558 chunks (15%)
Split Strategy	Recording-level (no leakage)

Table 11: Dataset Statistics

10.2 Training Configuration

Parameter	Value
Epochs	50 (early stopping)
Batch Size	32
Optimizer	AdamW
Learning Rate	0.0003
Weight Decay	0.05
Label Smoothing	0.1
Early Stop Patience	15 epochs
Gradient Clipping	1.0
Mixed Precision	FP16
Hardware	NVIDIA RTX 3050 (4GB)

Table 12: Training Configuration

10.3 Regularization

- **Dropout:** 50% before first dense layer, 30% before output

- **Weight Decay:** L2 penalty ($\lambda = 0.05$)
- **Label Smoothing:** Softens targets ($\epsilon = 0.1$)
- **Data Augmentation:** SpecAugment + Mixup
- **Early Stopping:** Stops when validation plateaus
- **Batch Normalization:** Normalizes layer inputs

11. Results & Analysis

11.1 Overall Performance

Metric	Validation	Test
Top-1 Accuracy	71.32%	67.42%
Top-3 Accuracy	79.78%	78.59%
Top-5 Accuracy	83.55%	82.57%
Macro F1	-	58.39%
Weighted F1	-	66.42%

Table 13: Performance Metrics

11.2 Performance by Sample Size

Category	Mean F1	Species
High (600+ samples)	0.682	11
Medium (300-599)	0.624	24
Low (100-299)	0.503	17
Very Low (<100)	0.256	2

Table 14: Performance by Sample Count

11.3 Best Performing Species

Species	English Name	F1
Myiarchus cinerascens	Ash-throated Flycatcher	0.939
Polioptila caerulea	Blue-gray Gnatcatcher	0.926
Corvus brachyrhynchos	American Crow	0.911
Selasphorus platycercus	Broad-tailed Hummingbird	0.860
Artemisiospiza belli	Bell's Sparrow	0.849

Table 15: Top 5 Species

11.4 Challenging Species

Species	English Name	F1	Issue
Bucephala albeola	Bufflehead	0.00	Few samples
Hirundo rustica	Barn Swallow	0.00	Confusion
Haliaeetus leucocephalus	Bald Eagle	0.11	Similar calls
Passerina caerulea	Blue Grosbeak	0.14	Confusion

Table 16: Challenging Species

12. Conclusions

This project successfully developed a deep learning system for multi-species bird identification from audio recordings.

Key Achievements:

- Implemented EfficientNet-B0 CNN with transfer learning
- Achieved 67.42% test accuracy on 54-class problem
- Reached 82.57% top-5 accuracy
- Reduced training time 5-10× via pre-computed spectrograms
- Fixed critical data leakage in original pipeline

Limitations:

- Two species achieve 0% F1 due to limited samples
- Class imbalance affects rare species performance
- Single-label classification limits multi-species detection

Future Work:

- Collect more data for underrepresented species
- Implement multi-label classification
- Add attention mechanisms for temporal localization
- Deploy as real-time monitoring application

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